Grover

Grover's Search Algorithm

A step-by-step chapter: from reflections in the plane to amplitude amplification

What you will be able to do after this chapter

- use the trigonometric identities and reflection matrices behind the algorithm without a calculator;
- manipulate superpositions, signs, measurements and multi-controlled gates with confidence;
- build the oracle and the diffusion circuit for any marked string, and count their gates;
- trace amplitudes by hand with inversion about the mean, and explain the same steps as a rotation;
- find the optimal number of iterations, and predict over-rotation, for one or for many solutions;
- reason about unknown M, exact and fixed-point search, amplitude amplification and quantum counting;
- solve every type of question in the accompanying 80-question Grover assessment.

Companion material in the same repository: `presentations/Grover_Lecture.pdf` (slides), `Grover.ipynb` (CUDA-Q notebook) and `assessments/` (question paper and answer book).

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Grover — Quantum Search

How to use this chapter

The chapter is built as a staircase. Sections 1–3 are the *mathematics* you need (trigonometry, reflections and rotations, averages and counting). Sections 4–5 are the *quantum mechanics* (states, measurement, gates, kickback). Sections 6–11 are *Grover's algorithm itself.* Sections 12–14 are the *advanced* material: unknown M, exact and fixed-point search, amplitude amplification, quantum counting and the limits of the speed-up. Section 15 collects problem-solving strategies and Section 16 has practice problems with answers.

Each section has: the idea in words, the precise statement, a derivation, **worked examples** solved step by step, **common traps** (the wrong answers examiners like to offer) and a short checklist. Do the worked examples with a pencil before reading the solution.

Definition: Conventions used everywhere in this chapter

- n qubits, $N = 2^n$ items, M marked items ($M = 1$ unless stated otherwise).
- $|x\rangle = |x_1 x_2 \dots x_n\rangle$ with $x = x_1 2^{n-1} + \cdots + x_n 2^0$: **qubit 1 is the most significant bit**.
- $|s\rangle = H^{\otimes n}|0\dots0\rangle = \frac{1}{\sqrt{N}}\sum_x |x\rangle$; oracle $U_\omega|x\rangle = (-1)^{f(x)}|x\rangle$; diffusion $D = 2|s\rangle\langle s| - I$; iteration $G = DU_\omega$ (oracle first).
- $\sin\theta = \sqrt{M/N}$, $P_k = \sin^2((2k+1)\theta)$, $k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right)$.
- **Notebook naming:** `Grover.ipynb` writes `N_QUBITS` (or N) for our n, $M = 2^N$ for our N, and O for our U_ω. It treats one marked item and prints qubit 0 first. The formulas are the same.

1 Complex phases and trigonometry

1.1 Complex numbers as phases

A complex number $z = a + ib$ has modulus $|z| = \sqrt{a^2+b^2}$ and polar form $z = |z|e^{i\varphi}$, where $e^{i\varphi} = \cos\varphi + i\sin\varphi$. Powers multiply the angle. *Example:* $\frac{1-i}{\sqrt{2}} = e^{-i\pi/4}$, so its cube is $e^{-3i\pi/4}$. The sign flip of the oracle is the phase $-1 = e^{i\pi}$: it changes direction, never length.

1.2 The identities you need

Everything about the success probability follows from the angle-addition formulas:

$$\cos 2\alpha = 1 - 2\sin^2\alpha = 2\cos^2\alpha - 1, \qquad \sin 2\alpha = 2\sin\alpha\cos\alpha, \qquad \sin 3\alpha = 3\sin\alpha - 4\sin^3\alpha.$$

The last one, squared and written with $s = \sin^2\alpha$, is worth remembering:

Key result: Probability after one iteration

$$\sin^2 3\alpha = s(3-4s)^2, \qquad s = \sin^2 \alpha.$$

Worked example 1.1: An exact value

Let $\sin\theta = \frac{1}{3}$. Find $\sin^2 3\theta$ and $\cos 2\theta$.

Step 1. $\sin 3\theta = 3\cdot\frac{1}{3} - 4\cdot\frac{1}{27} = 1 - \frac{4}{27} = \frac{23}{27}$, so $\sin^2 3\theta = \frac{529}{729}$.

Step 2. $\cos 2\theta = 1 - 2\cdot\frac{1}{9} = \frac{7}{9}$.

Check with the boxed form: $s = \frac{1}{9}$ gives $\frac{1}{9}\left(3-\frac{4}{9}\right)^2 = \frac{1}{9}\cdot\frac{529}{81} = \frac{529}{729}$. ✓

1.3 Small angles

For small x: $\sin x \approx x$ and $\arcsin x = x + \frac{x^3}{6} + \dots$. So $\arcsin x$ is slightly *larger* than x for $x > 0$. Example: $\arcsin 0.125 = 0.12533$. This is why $\theta \approx 1/\sqrt{N}$ is good for large N but not exact.

1.4 The curve $\sin^2((2k+1)\theta)$

The function $\sin^2$ has period π (not 2π) and equals 1 at odd multiples of $\frac{\pi}{2}$. Moving from k to $k+p$ adds $2p\theta$ to the angle.

Key result: Peak and period

$\sin^2((2k+1)\theta) = 1$ needs $(2k+1)\theta$ to be an odd multiple of $\frac{\pi}{2}$. The sequence repeats with period p whenever $2p\theta$ is a multiple of π; the natural period is $\frac{\pi}{2\theta}$ iterations when this is an integer.

Worked example 1.2: First maximum, period and window

Let $\theta = \frac{\pi}{14}$.

Step 1. First maximum: $(2k+1)\frac{\pi}{14} = \frac{\pi}{2}$ gives $2k+1 = 7$, so $k = 3$.

Step 2. Period: $2p\frac{\pi}{14} = m\pi$ gives $p = 7m$, so the smallest period is 7.

Step 3. Window $P \geq \frac{3}{4}$: the angle must be within $\frac{\pi}{6}$ of $\frac{\pi}{2}$, i.e. $\frac{\pi}{3} \leq (2k+1)\frac{\pi}{14} \leq \frac{2\pi}{3}$, so $4.67 \leq 2k+1 \leq 9.33$: $2k+1 \in \{5,7,9\}$, or $k = 2,3,4$.

Checklist 1. Square at the end; period of $\sin^2$ is π; use degrees if it makes the list of angles easier to read; check with $s(3-4s)^2$.

2 Reflections and rotations

2.1 Reflection matrices

Definition: Reflection about a unit vector

For a unit vector $|u\rangle$, the operator $R_u = 2\,|u\rangle\langle u| - I$ fixes $|u\rangle$ and negates every vector orthogonal to $|u\rangle$.

Because $\langle u|u\rangle = 1$, expanding gives $R_u^2 = 4\,|u\rangle\,\langle u|u\rangle\,\langle u| - 4\,|u\rangle\langle u| + I = I$. So a reflection is its own inverse; it is Hermitian, and hence unitary.

Key result: Spectrum and trace in dimension N

R_u has eigenvalue $+1$ once and -1 exactly $N-1$ times, so $\operatorname{tr} R_u = 2 - N$. The projector $|u\rangle\langle u|$ has trace 1 and is *not* unitary.

Worked example 2.1: A 2×2 reflection

Let $u = (\frac{5}{13}, \frac{12}{13})^T$.

Step 1. $uu^T = \frac{1}{169}\left(\begin{smallmatrix} 25 & 60 \\ 60 & 144 \end{smallmatrix}\right)$, so $2uu^T - I = \frac{1}{169}\left(\begin{smallmatrix} 50-169 & 120 \\ 120 & 288-169 \end{smallmatrix}\right) = \frac{1}{169}\left(\begin{smallmatrix} -119 & 120 \\ 120 & 119 \end{smallmatrix}\right)$.

Step 2. Check that it fixes u: first entry $\frac{1}{169 \cdot 13}(-119 \cdot 5 + 120 \cdot 12) = \frac{845}{2197} = \frac{5}{13}$. ✓ Trace 0, determinant -1.

2.2 Two reflections make a rotation

In the plane, a reflection in the line at angle α sends a vector at angle ϕ to angle $2\alpha - \phi$.

Key result: Composition

Reflecting in the line at α and then in the line at β gives $\phi \mapsto 2\beta - (2\alpha - \phi) = \phi + 2(\beta - \alpha)$: a **rotation by twice the angle between the lines**, directed from the first line to the second.

Worked example 2.2: Order matters

Lines at $20°$ and $50°$. First $20°$, then $50°$: rotation by $+60°$. In the opposite order: rotation by $-60°$. The product of two reflections has determinant $(-1)(-1) = +1$, so it can never be a reflection.

2.3 Eigenvalues and powers of a rotation

$R(\alpha) = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$ has eigenvalues $e^{\pm i\alpha}$ with eigenvectors $\frac{1}{\sqrt{2}}(1, \mp i)^T$, and powers simply add angles: $R(\alpha)^k = R(k\alpha)$.

Worked example 2.3: Eigenvalues and a power

Step 1. Let $\alpha = 2\theta$ with $\sin\theta = \frac{1}{3}$. Then $\cos 2\theta = \frac{7}{9}$ and $\sin 2\theta = 2 \cdot \frac{1}{3} \cdot \frac{2\sqrt{2}}{3} = \frac{4\sqrt{2}}{9}$. Eigenvalues: $\frac{7 \pm 4\sqrt{2}\,i}{9}$; modulus $\sqrt{49 + 32}/9 = 1$. ✓

Step 2. The rotation by 45° satisfies $R^k = -I$ first at $k = 4$, and $R^k = +I$ first at $k = 8$.

Common trap

Eigenvalues of a rotation by 2θ involve 2θ, not θ. Real numbers ± 1 occur only for rotations by 0 or π — and for reflections.

Checklist 2. Test a reflection by what it fixes; determinant -1 (plane) tells reflection from rotation; rotation angle is twice the angle between the lines.

3 Averages, counting and classical search

3.1 Inversion about the mean

Definition: Inversion about the mean

For a list $v_1, \ldots, v_N$ with mean $\bar{v}$: $v_j \mapsto 2\bar{v} - v_j$. Every entry is reflected through the mean: a distance d below becomes a distance d above.

The mean is unchanged and the sum of squares is preserved (Section 7 shows why this is unitary).

Worked example 3.1: Sixteen entries

v has 15 entries $+1$ and one entry -1. The mean is $\frac{14}{16} = \frac{7}{8}$. The special entry becomes $\frac{7}{4} + 1 = \frac{11}{4}$; the others become $\frac{7}{4} - 1 = \frac{3}{4}$. Length check: $15 \cdot \frac{9}{16} + \frac{121}{16} = \frac{256}{16} = 16$. ✓

3.2 Counting solutions

The quantity that controls Grover's algorithm is the **fraction of solutions** M/N. If the conditions involve disjoint groups of variables, the counts multiply. A useful consequence of the count: $\sum_x (-1)^{f(x)} = (N - M) - M = N - 2M$.

Worked example 3.2: A small Boolean function

$f = (x_1 \wedge x_2) \vee x_3$ on three bits. If $x_3 = 1$: all 4 choices of $x_1 x_2$ work. If $x_3 = 0$: only $x_1 x_2 = 11$. So $M = 5$, $M/N = \frac{5}{8}$, and $\sum_x (-1)^{f(x)} = 8 - 10 = -2$.

3.3 Classical search

Structure decides the cost. On a *sorted* list, binary search halves the range at each step, so $2^b - 1$ entries take at most b comparisons. On an *unsorted* list, a key that may be absent needs up to N comparisons. *Example:* $N = 255$: at most 8 and 255. Grover's algorithm concerns the unsorted case.

Key result: Expected number of queries

Drawing **without** replacement: $\dfrac{N+1}{M+1}$. (The M marked items split the $N - M$ others into $M + 1$ gaps of equal mean size.)

Drawing **with** replacement: $\frac{N}{M}$ (geometric distribution).
For one marked item: $\frac{N+1}{2}$ without replacement. The classical cost is $\Theta(N/M)$.

At least one success. r independent runs with success probability p succeed at least once with probability $1-(1-p)^r$. Example: $p=\frac{1}{2}$, $r=4$: $\frac{15}{16}$.

3.4 Square-root scaling

$\sqrt{N}=2^{n/2}$, so each extra qubit multiplies $\sqrt{N}$ by $\sqrt{2}$. Adding 4 bits multiplies the search space by 16 and $\sqrt{N}$ by 4.

Checklist 3. Mean first, then invert; check length; "at least one" means use the complement; with or without replacement gives different formulas.

4 Qubits, states and measurement

4.1 States and the Born rule

$|\psi\rangle=\sum_x c_x|x\rangle$ with $\sum_x |c_x|^2=1$. **Born rule:** measuring all qubits gives x with probability $|c_x|^2$. The uniform state $|s\rangle$ has every $c_x=1/\sqrt{N}$ and is a product of $|+\rangle$ states, so its qubits are independent. *Example:* in $H^{\otimes 4}|0000\rangle$, $P(x_1=0,x_4=1)=\frac{4}{16}=\frac{1}{4}$.

Method: Measuring only a property (subspace measurement)

(1) Group the basis states that have the property. (2) The probability is the sum of $|c_x|^2$ over the group. (3) The new state is that group alone, *re-normalised*; it stays a superposition.

Worked example 4.1: Grover's final step in miniature

$|\psi\rangle=\frac{1}{3}(2|00\rangle+|01\rangle+|10\rangle+\sqrt{3}|11\rangle)$. Ask "are the two bits equal?"

Step 1. The group is $\{|00\rangle,|11\rangle\}$. $P=\frac{4+3}{9}=\frac{7}{9}$.

Step 2. New state: $\frac{2|00\rangle+\sqrt{3}|11\rangle}{\sqrt{7}}$.

4.2 Global versus relative phase

Key result: Phases

$e^{i\gamma}|\psi\rangle$ and $|\psi\rangle$ are **the same physical state**. A sign on *one* amplitude of a superposition is a **relative** phase. It does not change computational-basis probabilities, but it does change results after an interfering gate such as H.

This single distinction explains both why the oracle alone gives nothing (Section 6) and why the -1 in the diffusion circuit is harmless (Section 10).

4.3 Normalisation and tensor products

Every term in the normalisation condition is a squared *modulus*: $|a+ib|^2 = a^2+b^2$. *Example:* $c(|0\rangle + (1-i)\,|1\rangle)$ needs $c^2(1+2)=1$, so $c=\frac{1}{\sqrt{3}}$. For two independent qubits, $(a_0\,|0\rangle + a_1\,|1\rangle) \otimes (b_0\,|0\rangle + b_1\,|1\rangle)$ has the amplitude $a_j b_k$ on $|jk\rangle$. *Example:* $\frac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle) \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ has $-\frac{i}{2}$ on $|11\rangle$.

4.4 Overlaps

$\langle a|b\rangle = \sum_x \overline{a_x} b_x$ (conjugate the *first* vector), and $|\,\langle a|b\rangle\,|^2$ is a probability. *Example:* for $a=\frac{1}{\sqrt{2}}(1,i)$, $b=(1,0)$: $\langle a|b\rangle = \frac{1}{\sqrt{2}}$; for $a=\frac{1}{\sqrt{2}}(1,i)$, $b=(0,1)$: $\langle a|b\rangle = -\frac{i}{\sqrt{2}}$. If $|r\rangle$ is the uniform superposition of the $N-1$ unmarked items, then $\langle r|s\rangle = (N-1)\frac{1}{\sqrt{N-1}\sqrt{N}} = \sqrt{\frac{N-1}{N}}$. For $N=8$: $\sqrt{7/8}$.

4.5 Memory of a state-vector simulation

2^n amplitudes × bytes per amplitude: 16 B in double precision (`fp64`), 8 B in single (`fp32`). $n=28$ in double: 4 GiB; $n=31$: 32 GiB. The notebook's iteration sweep holds two states at once and therefore needs twice this.

5 Gates for Grover

5.1 Hadamard layers

$H\,|0\rangle = |+\rangle$, $H\,|1\rangle = |-\rangle$, and $H^{\otimes n}\,|x\rangle = \frac{1}{\sqrt{N}}\sum_y (-1)^{x\cdot y}\,|y\rangle$ with $x\cdot y = \sum_j x_j y_j$. *Example:* in $H^{\otimes 3}\,|101\rangle$ the amplitude of $|110\rangle$ has $x\cdot y = 1+0+0 = 1$, so it is $-\frac{1}{\sqrt{8}}$.

	$\lvert 0\rangle$	$\lvert 1\rangle$	$\lvert +\rangle$	$\lvert -\rangle$
H	$\lvert +\rangle$	$\lvert -\rangle$	$\lvert 0\rangle$	$\lvert 1\rangle$
X	$\lvert 1\rangle$	$\lvert 0\rangle$	$\lvert +\rangle$	$-\lvert -\rangle$
Z	$\lvert 0\rangle$	$-\lvert 1\rangle$	$\lvert -\rangle$	$\lvert +\rangle$

Also $HXH = Z$, $HZH = X$. For one qubit, $2\,|+\rangle\langle +| - I = X$ and $2\,|0\rangle\langle 0| - I = Z$.

5.2 Multi-controlled gates

The multi-controlled Z (written $\mathrm{C}^{n-1}\mathrm{Z}$, or CCZ for three qubits) is $\mathrm{diag}(1,\ldots,1,-1) = I - 2\,|1\ldots1\rangle\langle 1\ldots1|$. It is symmetric: every qubit can be regarded as the target. Hadamards on the target turn a multi-controlled X into a multi-controlled Z, as in $(I\otimes I\otimes H)\,\mathrm{CCX}\,(I\otimes I\otimes H) = \mathrm{CCZ}$.

Key result: Moving the sign flip with X gates

Conjugating by X on selected qubits moves the affected basis state: if X_S maps $|a\rangle$ to $|1\ldots1\rangle$, then $X_S\,\mathrm{C}^{n-1}\mathrm{Z}\,X_S = I - 2\,|a\rangle\langle a|$.

Worked example 5.1: Selecting a basis state

To flip the sign of $|110\rangle$ on three qubits: apply X to qubit 3 (the only 0), then CCZ, then X on qubit 3 again. Only $|110\rangle$ is mapped to $|111\rangle$, and only it picks up the sign.

5.3 Phase kickback

Key result: Bit oracle to phase oracle

For $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$: $\quad U_f|x\rangle|-\rangle = (-1)^{f(x)}|x\rangle|-\rangle$, $\quad$ but $\quad U_f|x\rangle|+\rangle = |x\rangle|+\rangle$.

Reason: $|\pm\rangle$ are eigenvectors of X with eigenvalues ± 1; if $f(x) = 1$ the ancilla receives an X. The ancilla stays in a product with the input register, so it can be measured or re-used without disturbing anything.

Worked example 5.2: Toffoli as a two-qubit phase oracle

Start with $|00\rangle|1\rangle$ and apply H to all three, then Toffoli. With $f = x_1 \wedge x_2$: state $= \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle)|-\rangle$. Measuring the ancilla gives 0 or 1 with probability $\frac{1}{2}$ each and leaves the first factor unchanged.

5.4 Entanglement test

Key result: Product test for two qubits

$\sum c_{jk}|jk\rangle$ is a product state $\iff c_{00}c_{11} = c_{01}c_{10}$.

Examples: $\frac{1}{2}(1,-1,1,-1)$: $-\frac{1}{4} = -\frac{1}{4}$, product $|+\rangle|-\rangle$. $\quad \frac{1}{2}(1,1,-1,1)$: $\frac{1}{4} \neq -\frac{1}{4}$, entangled. So a single sign flip on $|s\rangle$ with $n = 2$ produces entanglement.

Common trap

$Z \otimes Z \otimes Z$ is not CCZ: it flips the sign of every odd-weight string. And $|-\rangle$, not $|+\rangle$, is the ancilla that gives kickback.

6 The search problem and the oracle

Definition: Unstructured search

Given a black box for $f : \{0,1\}^n \to \{0,1\}$ with $f(x) = 1$ exactly on the M marked items, find one marked item using as few queries as possible.

The oracle does not "know the answer" in a useful way: it only recognises a solution, as a program checks a candidate. For $M = 1$:

$$U_\omega = I - 2|x^*\rangle\langle x^*|.$$

Key result: The oracle alone reveals nothing

U_ω only changes signs. On $|s\rangle$ it leaves every magnitude at $1/\sqrt{N}$, so a measurement straight after the oracle is uniformly random.

Key result: The marked item does not matter

Only $\langle x^*|s\rangle = 1/\sqrt{N}$ enters the analysis, and it is the same for every x^*. This symmetry of $|s\rangle$ — not the fact that U_ω is diagonal — is why P_k is independent of the choice of x^*.

7 Diffusion: inversion about the mean

Definition: Diffusion operator

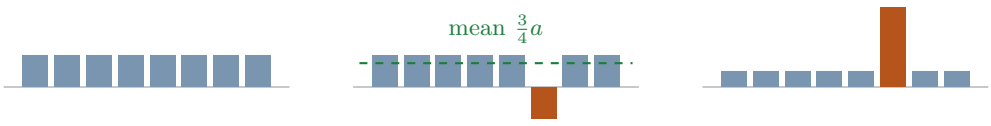
$D = 2\,|s\rangle\langle s| - I$. Since $\langle s|\psi\rangle = \frac{1}{\sqrt{N}}\sum_x c_x = \sqrt{N}\,\bar{c}$, the action on amplitudes is

$$c_x \longmapsto 2\bar{c} - c_x.$$

Its matrix has diagonal entries $\frac{2}{N} - 1$ and off-diagonal entries $\frac{2}{N}$, because every entry of $|s\rangle\langle s|$ is $\frac{1}{N}$. For $N = 8$ these are $-\frac{3}{4}$ and $\frac{1}{4}$. So D is exactly inversion about the mean, and it is unitary because it is a reflection. Also $D\,|s\rangle = |s\rangle$: if nothing is marked, nothing changes.

Method: Tracing amplitudes by hand

(1) Oracle: change the sign of the marked amplitudes. (2) Compute the mean of *all* N amplitudes. (3) Replace each c_x by $2\bar{c} - c_x$. (4) Check $\sum |c_x|^2 = 1$.



start: $|s\rangle$ after the oracle U_ω after the diffusion D

One iteration for $N = 8$, $|x^*\rangle = |101\rangle$, $a = 1/\sqrt{8}$: amplitudes $a \to \pm a \to (\frac{1}{2}a,\ \frac{5}{2}a)$.

Worked example 7.1: The smallest case, $N = 4$

Amplitudes $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Oracle on the last: $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, -\frac{1}{2})$; mean $\frac{1}{4}$. Inversion: $(0, 0, 0, 1)$. One iteration finds the item with **certainty**.

Worked example 7.2: Two iterations for $N = 32$

Let $a = 1/\sqrt{32}$.

Step 1. Oracle: mean $= \frac{30}{32}a = \frac{15}{16}a$. Marked: $\frac{15}{8}a + a = \frac{23}{8}a$. Unmarked: $\frac{15}{8}a - a = \frac{7}{8}a$. $P_1 = \frac{529}{64}a^2 = \frac{529}{2048} \approx 0.258$.

Step 2. Oracle again: marked $-\frac{23}{8}a$. Mean $= \frac{31\cdot\frac{7}{8} - \frac{23}{8}}{32}a = \frac{194}{256}a = \frac{97}{128}a$. Marked: $\frac{97}{64}a + \frac{184}{64}a = \frac{281}{64}a$. $P_2 = \frac{281^2}{64^2\cdot 32} \approx 0.602$.

Check: $\theta = 0.17771$: $\sin^2 3\theta = 0.258$ and $\sin^2 5\theta = 0.602$. ✓

Common trap

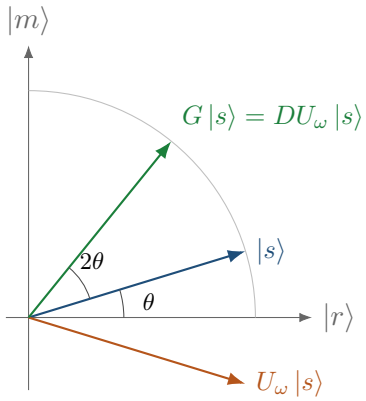
Compute the mean *after* the oracle, and include the marked item. Using the mean a of the starting state, or leaving the marked item out, gives wrong numbers that look plausible. Also do not stop at the amplitude: the probability is its square.

8 The geometric picture

Let $|m\rangle$ be the uniform superposition of the marked items and $|r\rangle$ that of the unmarked items. They are orthonormal and

$$|s\rangle = \cos\theta\ |r\rangle + \sin\theta\ |m\rangle\,, \qquad \sin\theta = \sqrt{M/N}.$$

Both O and D map this plane to itself, so the whole algorithm takes place in two dimensions — however large N is.



U_ω reflects in the axis $|r\rangle$; D reflects in the line through $|s\rangle$; together they rotate by 2θ towards $|m\rangle$.

Step 1. U_ω negates $|m\rangle$ and fixes $|r\rangle$: it is the **reflection in the line through** $|r\rangle$ (angle 0).

Step 2. D fixes $|s\rangle$ and negates the perpendicular direction: it is the **reflection in the line through** $|s\rangle$ (angle θ).

Step 3. By Section 2, reflecting first at angle 0 and then at angle θ is a **rotation by** 2θ.

Step 4. $|s\rangle$ starts at angle θ from $|r\rangle$, so after k iterations the angle is $(2k+1)\theta$.

Key result: The state after k iterations

$$G^k\,|s\rangle = \sin\left((2k+1)\theta\right)|m\rangle + \cos\left((2k+1)\theta\right)|r\rangle\,, \qquad P_k = \sin^2\left((2k+1)\theta\right).$$

In the basis $\{|r\rangle\,, |m\rangle\}$: $G = \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix}$.

9 Success probability and the optimal number of iterations

We want $(2k+1)\theta$ as close as possible to $\frac{\pi}{2}$:

Key result: Optimal number of iterations

$$k_{\text{opt}} = \text{round}\left(\frac{\pi}{4\theta} - \frac{1}{2}\right) \approx \frac{\pi}{4}\sqrt{\frac{N}{M}}, \qquad P_{k_{\text{opt}}} \geq 1 - \sin^2\theta = 1 - \frac{M}{N}.$$

Bound: rounding leaves $(2k+1)\theta$ within θ of $\frac{\pi}{2}$, and $\sin^2(\frac{\pi}{2} \pm \theta) = \cos^2\theta$.

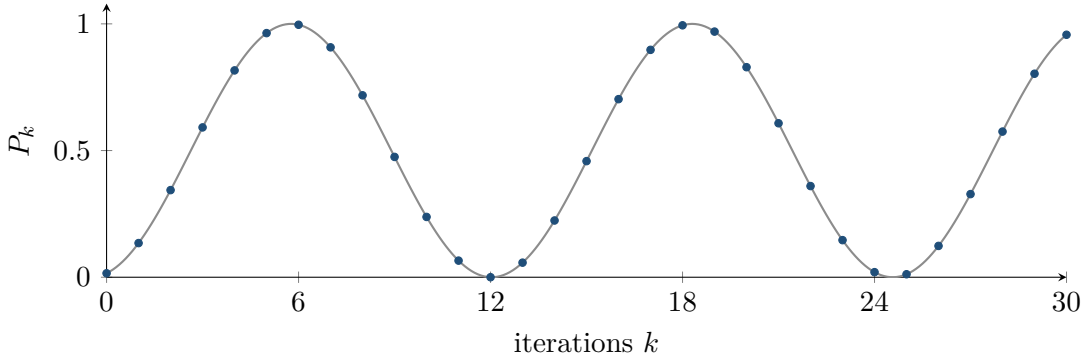
n	N	θ (rad)	k_{opt}	P
2	4	0.5236	1	1.0000
3	8	0.3614	2	0.9453
4	16	0.2527	3	0.9613
5	32	0.1777	4	0.9992
6	64	0.1253	6	0.9966
8	256	0.0625	12	0.9999
10	1024	0.0313	25	0.9995
15	32 768	0.0055	142	1.0000
20	1 048 576	0.00098	804	1.0000

Worked example 9.1: Why $-\frac{1}{2}$ matters: $n = 8$

Step 1. Rule of thumb: $\frac{\pi}{4}\sqrt{256} = 12.57$, which would round to 13.

Step 2. Exact rule: $\frac{\pi}{4\theta} - \frac{1}{2} = 12.56 - 0.5 = 12.06$, which rounds to 12.

Step 3. Compare: $\sin^2 25\theta = 0.99995$ but $\sin^2 27\theta = 0.986$. The rule of thumb gives the wrong integer here.



P_k for $n = 6$, one marked item: first peak at $k = 6$, nearly zero near $k = 12$ to 13, and then a second rise. The notebook measures this same shape in its own Section 9.

Key result: Over-rotation

More iterations are not better. Past k_{opt} the state rotates beyond $|m\rangle$ and P_k falls. At $k \approx 2k_{\text{opt}}$ the state is close to $\pm|r\rangle$ and the marked item is *less* likely than for a random guess. The curve is periodic: one full period is $\frac{\pi}{2\theta}$ iterations.

Worked example 9.2: Size of the speed-up

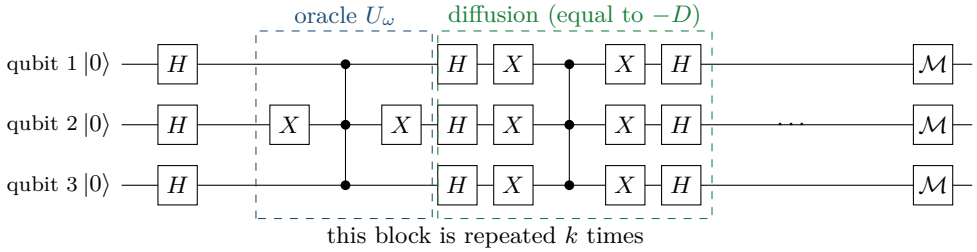
$n = 30$: classical average $2^{29} \approx 5.4 \times 10^8$ queries; Grover $\frac{\pi}{4} \cdot 2^{15} \approx 25\,736$ iterations; ratio $\frac{2}{\pi}\sqrt{N} \approx 20\,861$. Large, but only quadratic.

Worked example 9.3: Shot statistics

With $P = 0.99$ and $S = 1000$ shots: marked count $= SP = 990$, with standard deviation $\sqrt{SP(1-P)} = \sqrt{9.9} \approx 3.1$. The Poisson rule $\sqrt{990} \approx 31$ is wrong here because P is close to 1. All S shots succeed with probability $P^S \approx e^{-S(1-P)}$: for $P = 0.9999$ and $S = 1000$, about 0.905. A perfect count does not prove $P = 1$.

10 The Grover circuit

10.1 Building it



Grover circuit for $n = 3$, $|x^*\rangle = |101\rangle$. A column of three dots is the multi-controlled Z (symmetric, so no target is drawn).

Method: Oracle for one marked string

Put X on every qubit where x^* has a 0, apply the multi-controlled Z, and repeat the same X gates. *Example:* $x^* = 0110$ uses X on qubits 1 and 4.

10.2 The diffusion circuit is $-D$

Step by step, using the rule for moving a reflection with a unitary:

$$\mathrm{C}^{n-1}\mathrm{Z} = I - 2\,|1\ldots1\rangle\langle1\ldots1| \xrightarrow{X^{\otimes n}} I - 2\,|0\ldots0\rangle\langle0\ldots0| \xrightarrow{H^{\otimes n}} I - 2\,|s\rangle\langle s| = -D.$$

The factor -1 is a global phase, so the circuit is correct. The notebook's kernel has the same remark in its code comment.

10.3 Counting gates

Key result: Gates per iteration

With z the number of 0s in x^*: oracle $= 2z + 1$; diffusion $= 4n + 1$; so total $= 4n + 2z + 2$. Whole circuit: $n + k(4n + 2z + 2)$ gates, plus n measurements.

Worked example 10.1: Two counts

Step 1. $n = 5$, $x^* = 10010$: $z = 3$, so $20 + 6 + 2 = 28$ gates per iteration.

Step 2. $n = 4$, $x^* = 1011$, $k = k_{\text{opt}} = 3$: $z = 1$, $16 + 2 + 2 = 20$ per iteration; total $4 + 3 \cdot 20 = 64$ gates.

10.4 Cost of a multi-controlled gate

A multi-controlled Z with m controls can be built with $m-1$ clean ancillas, $2(m-1)$ Toffoli gates and one CZ. *Example:* $n=6$ means $m=5$: 8 Toffolis per gate, so 16 per iteration. This is why hardware circuits grow quickly when multi-controlled gates are decomposed.

10.5 Reading the notebook's numbers

`Grover.ipynb` has one marked item, no ancillas, and calls `z.ctrl` directly. Its output can be decoded with the formulas above.

- **Gate statistics.** `h` $= n+2nk$, `x` $= k(2n+2z)$, multi-controlled $Z = 2k$. *Example with* $n=10$, $k=25$, $z=4$*:* 510, 700 and 50.
- **Memory.** On the CPU with the sweep on, it estimates $2 \times 2^n \times 16$ bytes, since two states are held at once. *Example:* $n=20$ gives 32 MiB.
- **Sweep points.** `stride` $= \lceil$`sweep_max/SWEEP_MAX_POINTS`$\rceil$ with `sweep_max` $= 2k_{\text{opt}}+1$, and the number of points is $1+\lfloor$`sweep_max/stride`$\rfloor$. *Example:* $k_{\text{opt}}=25$ with 60 allowed points: `sweep_max` $= 51$, stride 1, and 52 points.

11 Several solutions

Nothing changes except the angle: $\sin\theta = \sqrt{M/N}$, $k_{\text{opt}} \approx \frac{\pi}{4}\sqrt{N/M}$. The final state is the uniform superposition of the marked items, so measuring returns a uniformly random solution.

Worked example 11.1: $N=1024$, $M=4$

$\sin\theta = \frac{2}{32} = \frac{1}{16}$, $\theta = 0.06254$, $\frac{\pi}{4\theta} - \frac{1}{2} = 12.06$, so $k=12$. Compared with $M=1$ (where $k=25$), the work has halved, because $\sqrt{M}=2$.

M/N	θ	$P_1 = \sin^2 3\theta$	What happens
$\frac{1}{8}$	20.7°	$\frac{25}{32}$	normal; one more step gives 0.945
$\frac{1}{4}$	30°	1	**one iteration is exact**
$\frac{1}{2}$	45°	$\frac{1}{2}$	**flat**: $P_k = \frac{1}{2}$ for every k
$\frac{3}{4}$	60°	0	**worst case**: one iteration erases all marked weight

Common trap

When $M > N/2$ do not iterate: measuring immediately succeeds with probability $M/N > \frac{1}{2}$. Iterating can make things much worse. The formula for k_{opt} agrees: for $M > N/2$ we have $\theta > \frac{\pi}{4}$, so $\frac{\pi}{4\theta} - \frac{1}{2} < \frac{1}{2}$ and it rounds to 0.

12 Unknown M, exact search and fixed-point search

12.1 Unknown M (BBHT)

Choosing k needs θ, hence M. Boyer, Brassard, Høyer and Tapp avoid this with randomness.

Key result: Average over a random k

$$\frac{1}{m}\sum_{k=0}^{m-1}\sin^2\left((2k+1)\theta\right) = \frac{1}{2} - \frac{\sin 4m\theta}{4m\sin 2\theta},$$

which is at least $\frac{1}{4}$ as soon as $m \geq \frac{1}{\sin 2\theta} \approx \frac{1}{2}\sqrt{N/M}$.

Method: BBHT

(1) Set $m = 1$. (2) Choose k uniformly from $\{0, \ldots, \lceil m\rceil - 1\}$ and run k iterations. (3) Measure and test the result with the oracle; stop on success. (4) Otherwise set $m \leftarrow \min(\lambda m, \sqrt{N})$ with λ such as $\frac{6}{5}$, and return to (2). The expected number of queries is $O(\sqrt{N/M})$.

Example: $n = 10$, $M = 1$: threshold $\approx \frac{1}{2}\sqrt{1024} = 16$.

12.2 Exact search

If $(2k+1)\theta$ misses $\frac{\pi}{2}$, shrink the angle a little: use $k = \lceil \frac{\pi}{4\theta} - \frac{1}{2}\rceil$ and $\theta' = \frac{\pi}{2(2k+1)} \leq \theta$. A smaller marked weight (for example from an extra ancilla rotation) gives $P = 1$. *Example:* $N = 8$: $\theta = 0.3614$, ceiling of 1.67 is $k = 2$, and $\theta' = \frac{\pi}{10} = 0.3142 < \theta$. ✓ You must round *up*: rounding down needs a larger angle, which reducing the marked weight cannot provide.

12.3 Fixed-point search

Grover's $\pi/3$ method replaces the sign flips by phases $\frac{\pi}{3}$. One recursion step maps the failure probability $\varepsilon \mapsto \varepsilon^3$, so $\varepsilon_j = \varepsilon^{3^j}$; extra steps cannot over-rotate. *Example:* $\varepsilon = 0.2 \to 0.008 \to 5.12 \times 10^{-7}$. The basic form gives up the quadratic speed-up; later variants recover it.

12.4 Maximum success is not minimum cost

If you repeat until success, with a free check of each result, the expected cost is k/P_k queries.

Worked example 12.1: $N = 16$

$P_1 = 0.473$, $P_2 = 0.908$, $P_3 = 0.961$. Expected cost: $k/P_k = 2.12, 2.20, 3.12$. So $k = 1$ is cheapest, even though $k_{\text{opt}} = 3$. If each check costs one extra query the cost becomes $(k+1)/P_k = 4.23, 3.30, 4.16$, and $k = 2$ wins. Read the question to see which is asked.

13 Amplitude amplification and quantum counting

13.1 Amplitude amplification

Replace $H^{\otimes n}$ by any procedure $\mathcal{A}$ (no measurement) with success probability a:

Key result: Amplification

$Q = -\mathcal{A}S_0\mathcal{A}^{-1}S_\chi$, with S_χ flipping the sign of good states and $S_0 = I - 2\,|0\rangle\langle 0|$. With $\sin^2\theta = a$: $P_k = \sin^2((2k+1)\theta)$, using $\mathcal{A}$ or $\mathcal{A}^{-1}$ in total $\mathbf{2k+1}$ times. Grover is the case $\mathcal{A} = H^{\otimes n}$, $a = M/N$.

Worked example 13.1: $a = 0.04$

$\theta = \arcsin 0.2 = 0.2014$, $\frac{\pi}{4\theta} - \frac{1}{2} = 3.40$, so $k = 3$; 7 uses of $\mathcal{A}$; $P = \sin^2 7\theta = 0.974$. Classical repetition needs $1/a = 25$ runs on average.

13.2 Eigenstructure of G

Since G is a rotation by 2θ on the plane (Section 2), its eigenvalues there are $e^{\pm 2i\theta}$ with eigenvectors $\frac{1}{\sqrt{2}}(|r\rangle \mp i\,|m\rangle)$. The start state $|s\rangle$ is an equal-weight superposition of the two.

13.3 Quantum counting

Apply phase estimation to G with start state $|s\rangle$. Writing the eigenvalue as $e^{2\pi i\varphi}$ gives $\varphi = \frac{\theta}{\pi}$ or $1 - \frac{\theta}{\pi}$. So outcome y on t counting qubits gives

$$\tilde{\theta} = \frac{\pi y}{2^t}, \qquad \tilde{M} = N \sin^2 \tilde{\theta}.$$

Outcomes y and $2^t - y$ give the same $\tilde{M}$.

Worked example 13.2: $N = 32$, $M = 8$, $t = 4$

Step 1. $\sin^2\theta = \frac{1}{4}$, $\theta = \frac{\pi}{6}$, so $\varphi = \frac{1}{6}$ and $\frac{5}{6}$.

Step 2. On the grid: $16 \cdot \frac{1}{6} = 2.67$ and $16 \cdot \frac{5}{6} = 13.33$, so the likely outcomes are $y = 3$ and $y = 13$.

Step 3. From $y = 3$: $\tilde{M} = 32 \sin^2 \frac{3\pi}{16} = 32 \times 0.3087 \approx 9.9$. The true value is 8: the estimate improves with every extra counting qubit.

Common trap

The eigenphase is θ/π, not $\theta/(2\pi)$: the eigenvalue is $e^{2i\theta}$, so $2\pi\varphi = 2\theta$.

14 Limits and applications

Key result: Optimality

Every quantum algorithm with black-box access needs $\Omega(\sqrt{N/M})$ queries (Bennett, Bernstein, Brassard, Vazirani). Zalka proved that $\frac{\pi}{4}\sqrt{N}$ is optimal for $M = 1$ including the constant. **Grover cannot be improved in the black-box setting.**

- **NP-complete problems:** search over 2^n assignments takes about $2^{n/2}$ iterations, which is still exponential. Grover does not make these problems polynomial. For 40 variables, $\frac{\pi}{4}2^{20} \approx 823\,550$ iterations.
- **Parallel search:** p machines reduce the time by only $\sqrt{p}$, and the total queries increase by $\sqrt{p}$. Four machines: half the time, twice the queries. Classically, the time falls by p with constant total work.
- **Keys:** a k-bit key needs about $2^{k/2}$ iterations, and each one must run the cipher reversibly. Doubling the key length restores the margin: AES-256 still has about 2^{128} iterations.

- **Noise:** in a simple model, $P_k \approx f^k \sin^2((2k+1)\theta) + (1-f^k)/N$. For $f = 0.995$, $k = 25$: $f^k \approx 0.88$. Because $k \propto \sqrt{N}$, large searches need very low error rates.
- **Simulation:** state-vector work grows like $k \cdot 2^n \sim 2^{1.5n}$, a factor of $2\sqrt{2}$ per qubit. The notebook's $n = 15$ default takes about a minute including its sweep.

Task	Queries	Comment
Unstructured search, classical	$\Theta(N)$	must check a constant fraction of items
Unstructured search, quantum	$\Theta(\sqrt{N})$	Grover; optimal
Collision in a 2-to-1 function	$\Theta(N^{1/3})$	Brassard–Høyer–Tapp; lower bound by Aaronson–Shi
Element distinctness	$\Theta(N^{2/3})$	Ambainis's quantum walk

Other uses include minimum finding (Dürr–Høyer) and speeding up the classical part of many algorithms by amplitude amplification.

15 Problem-solving strategies

15.1 Strategy by question format

- **Single-correct numerical:** compute fully, then sanity-check (probabilities in $[0, 1]$; $\sum |c_x|^2 = 1$; compare inversion about the mean with $\sin^2((2k+1)\theta)$). Many distractors are "forgot to square", "used $\sqrt{N}$ without $\frac{\pi}{4}$", "dropped the $-\frac{1}{2}$", "used the mean instead of the new amplitude".
- **Matching lists:** solve the easiest row first; often one row fixes the option.
- **Assertion–Reason:** decide A and R *independently* first. Only if both are true ask: does R *cause* A?
- **Statement I/II:** judge each statement alone; one counterexample kills "never" or "always" ($N = 4$, $k = 2$ is a useful one).
- **Paragraph:** the first question usually sets up a quantity reused in the second; write it down clearly.

15.2 Fast checks that catch most mistakes

1. Is the state still normalised after your step?
2. Is M/N a special ratio ($\frac{1}{4}, \frac{1}{2}, \frac{3}{4}$)?
3. Compute P_k both ways: amplitudes and $\sin^2((2k+1)\theta)$.
4. Is k past k_{opt}? Then expect over-rotation.
5. Does the X pattern match the *zeros* of x^*?

15.3 Map: assessment questions → sections of this chapter

Questions	Where to learn it
Q.1, Q.5, Q.15, Q.17, Q.19, Q.20	Section 1 (phases, identities, the $\sin^2$ curve, peak and period)
Q.2–Q.4, Q.16, Q.18, Q.30	Section 2 (reflections, rotations, eigenvalues, traces)
Q.7–Q.11, Q.13	Section 3 (inversion about the mean, counting, classical search, scaling)
Q.12, Q.14, Q.21, Q.22, Q.28, Q.29, Q.31, Q.32, Q.38	Section 4 (normalisation, tensor products, Born rule, phases, subspace measurement, overlaps, memory)
Q.23–Q.27, Q.33, Q.35, Q.36, Q.39, Q.40	Section 5 (Hadamard layers, X conjugation, kickback, entanglement)
Q.37, Q.57	Section 6 (what the oracle does and does not do)
Q.34, Q.41, Q.42, Q.44, Q.48, Q.59, Q.60	Section 7 (the matrix of D, tracing amplitudes by hand)
Q.6, Q.43, Q.52–Q.55, Q.58, Q.73	Section 9 (optimal k, over-rotation, speed-up, shots)
Q.45–Q.47, Q.56, Q.65, Q.67, Q.70, Q.72	Section 10 (circuit, $-D$, gate counts, reading the notebook's numbers)
Q.49–Q.51, Q.61, Q.78	Section 11 (several solutions, special ratios)
Q.62, Q.63, Q.74	Section 12 (BBHT, exact search, cost)
Q.66, Q.76, Q.79, Q.80	Section 13 (amplification, counting)
Q.64, Q.68, Q.69, Q.71, Q.75, Q.77	Section 14 (optimality, keys, noise, simulation, complexity)

(Section 8, the geometric picture, underlies all of Q.41–Q.80.)

16 Practice problems

Try each before looking at the answers on the next page. They use the same ideas as the assessment but different numbers.

P1. Let $\sin\theta = \frac{1}{5}$. Find $\sin^2 3\theta$ exactly.

P2. Find $2uu^T - I$ for $u = \frac{1}{\sqrt{2}}(1,1)^T$.

P3. Lines at 15° and 75°: reflect in the first, then in the second. What is the combined map?

P4. $v = (1,1,1,-1)$. Find the result of inversion about the mean.

P5. $N = 15$ items, $M = 4$ marked. Expected draws without replacement until the first marked one?

P6. $f = x_1 \wedge (x_2 \vee x_3)$ on three bits. Find M/N.

P7. $|\psi\rangle = \frac{1}{\sqrt{6}}(|00\rangle + |01\rangle - 2\,|11\rangle)$. Find P(the bits are equal) and the state afterwards.

P8. Which basis state has its sign flipped by $X_1X_3\,\mathrm{CCZ}\,X_1X_3$?

P9. Is $\frac{1}{2}(|00\rangle - |01\rangle - |10\rangle - |11\rangle)$ entangled?

P10. $N = 64$, one marked item, one iteration: find the marked amplitude (in units of $a = 1/8$) and P_1.

P11. Find k_{opt} for $n = 12$, one marked item ($\theta \approx 0.015626$).

P12. $N = 64$, $M = 16$: find P_1.

P13. Gates per iteration for $n = 7$, $x^* = 0011010$.

P14. $N = 128$, $M = 2$: find k_{opt}.

P15. Amplitude amplification with $a = \frac{1}{9}$ ($\theta \approx 0.3398$): find the optimal k, the uses of $\mathcal{A}$, and P.

P16. Quantum counting with $N = 16$, $M = 4$, $t = 4$: the likely y and the resulting $\tilde{M}$.

Answers to the practice problems

P1. $\sin 3\theta = \frac{3}{5} - \frac{4}{125} = \frac{71}{125}$; so $\sin^2 3\theta = \frac{5041}{15625}$.

P2. $uu^T = \frac{1}{2}\left(\begin{smallmatrix}1&1\\1&1\end{smallmatrix}\right)$, so $2uu^T - I = \left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right) = X$.

P3. Rotation by $+120°$.

P4. Mean $\frac{1}{2}$; result $(0, 0, 0, 2)$. Length check: $4 = 4$.

P5. $\frac{16}{5} = 3.2$.

P6. $x_1 = 1$ and 3 of the 4 pairs (x_2, x_3): $M = 3$, so $\frac{3}{8}$.

P7. $P = \frac{1+4}{6} = \frac{5}{6}$; new state $\frac{|00\rangle - 2|11\rangle}{\sqrt{5}}$.

P8. $|010\rangle$.

P9. $c_{00}c_{11} = -\frac{1}{4}$ and $c_{01}c_{10} = \frac{1}{4}$: entangled.

P10. Mean $\frac{62}{64}a = \frac{31}{32}a$; marked $\frac{31}{16}a + a = \frac{47}{16}a$; $P_1 = \frac{2209}{256} \cdot \frac{1}{64} = \frac{2209}{16384} \approx 0.135$.

P11. $\frac{\pi}{4\theta} - \frac{1}{2} = 50.26 - 0.5 = 49.76$, so $k_{\text{opt}} = 50$.

P12. $M/N = \frac{1}{4}$, so $\theta = 30°$ and $P_1 = 1$.

P13. $z = 4$: $4 \cdot 7 + 8 + 2 = 38$.

P14. $\sin\theta = \frac{1}{8}$, $\theta = 0.12533$, $\frac{\pi}{4\theta} - \frac{1}{2} = 5.77$: $k_{\text{opt}} = 6$.

P15. $\frac{\pi}{4\theta} - \frac{1}{2} = 1.81$: $k = 2$; 5 uses; $P = \sin^2 5\theta \approx 0.983$.

P16. $\theta = \frac{\pi}{6}$, $\varphi = \frac{1}{6}, \frac{5}{6}$; $y \approx 2.67$ and 13.33, so $y = 3$ or 13; $\tilde{M} = 16\sin^2\frac{3\pi}{16} \approx 4.9$.

Summary of the chapter in ten lines

1. Classical search costs about N/M queries; Grover costs about $\frac{\pi}{4}\sqrt{N/M}$ iterations.
2. Oracle U_ω flips the sign of marked items: invisible until interference follows.
3. Diffusion $D = 2\,|s\rangle\langle s| - I$ is inversion about the mean: $c_x \mapsto 2\bar{c} - c_x$.
4. Circuit: X on the zeros of x^*, multi-controlled Z, X again; then H, X, multi-controlled Z, X, H (which is $-D$).
5. Two reflections are a rotation by twice the angle between them: G rotates by 2θ.
6. $\sin\theta = \sqrt{M/N}$; $P_k = \sin^2((2k+1)\theta)$; $k_{\text{opt}} = \text{round}(\frac{\pi}{4\theta} - \frac{1}{2})$.
7. More iterations over-rotate: P_k oscillates. $M = \frac{N}{4}$ is exact, $\frac{N}{2}$ is flat, $\frac{3N}{4}$ is the worst case for one iteration.
8. Unknown M: BBHT. Exact: shrink the angle. No over-rotation: fixed-point search.
9. Amplitude amplification: $\mathcal{A}$ with success a, $2k+1$ uses. Counting: phase estimation of G, whose eigenvalues are $e^{\pm 2i\theta}$.
10. $\Omega(\sqrt{N})$ is a proven lower bound: the speed-up is quadratic, never exponential.